ME101: Engineering Mechanics (3 1 0 8)

2023-2024 (II Semester)
Division II & IV

Instructor

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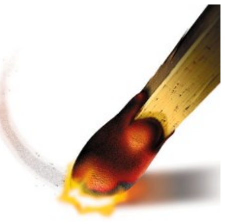
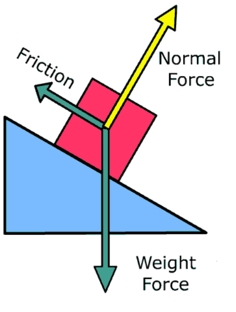
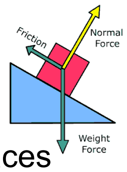
Friction

The effective design of a brake system, such as the one for this bicycle, requires an efficient capacity for the mechanism to resist frictional forces. In this chapter, we will study the nature of friction and show how these forces are considered in engineering analysis and design.





Friction



Usual Assumption till now:

Forces of action and reaction between contacting surfaces act normal to the surface

→ valid for interaction between smooth surfaces

→ often involves only a relatively small error in solution

→ in many cases ability of contacting surfaces to support tangential forces is very important (Ex: Figure above)

Frictional Forces

Tangential forces generated between contacting surfaces

- occur in the interaction between all real surfaces
- always act in a direction opposite to the direction of motion

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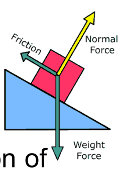
Friction



Frictional forces are Not Desired in some cases:

- Bearings, power screws, gears, flow of fluids in pipes, propulsion of aircraft and missiles through the atmosphere, etc
  - Friction often results in a loss of energy, which is dissipated in the form of heat
  - Friction causes Wear

Frictional forces are Desired in some cases:

- Brakes, clutches, belt drives, wedges
- walking depends on friction between the shoe and the ground

Ideal Machine/Process: Friction small enough to be neglected

Real Machine/Process: Friction must be taken into account

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Types of Friction

Dry Friction (Coulomb Friction)

occurs between unlubricated surfaces of two solids

Effects of dry friction acting on exterior surfaces of rigid bodies → ME101

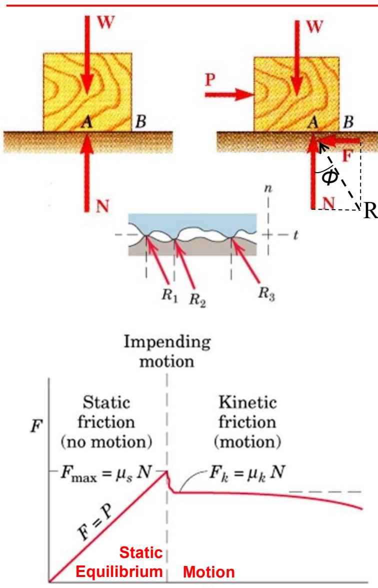
Fluid Friction

occurs when adjacent layers in a fluid (liquid or gas) move at a different velocities. Fluid friction also depends on viscosity of the fluid. → Fluid Mechanics

Internal Friction

occurs in all solid materials subjected to cyclic loading, especially in those materials, which have low limits of elasticity → Material Science

Mechanism of Dry Friction



- Block of weight *W* placed on horizontal surface. Forces acting on block are its weight and reaction of surface *N*.
- Small horizontal force *P* applied to block. For block to remain stationary, in equilibrium, a horizontal component *F* of the surface reaction is required. *F* is a *Static-Friction force*.
- As *P* increases, static-friction force *F* increases as well until it reaches a maximum value F_m.

$$F_m = \mu_s N$$

- Further increase in P causes the block to begin to move as *F* drops to a smaller *Kinetic-Friction force* F_k.

$$F_k = \mu_k N$$

μ_s is the Coefficient of Static Friction
μ_k is the Coefficient of Kinetic Friction

Mechanism of Dry Friction

Table 8.1. Approximate Values of Coefficient of Static Friction for Dry Surfaces

Metal on metal	0.15–0.60
Metal on wood	0.20–0.60
Metal on stone	0.30–0.70
Metal on leather	0.30–0.60
Wood on wood	0.25–0.50
Wood on leather	0.25–0.50
Stone on stone	0.40–0.70
Earth on earth	0.20–1.00
Rubber on concrete	0.60–0.90

A friction coefficient reflects roughness, which is a geometric property of surfaces

- Maximum static-friction force:

$$F_m = \mu_s N$$

- Kinetic-friction force:

$$F_k = \mu_k N$$
$$\mu_k \cong 0.75\mu_s$$

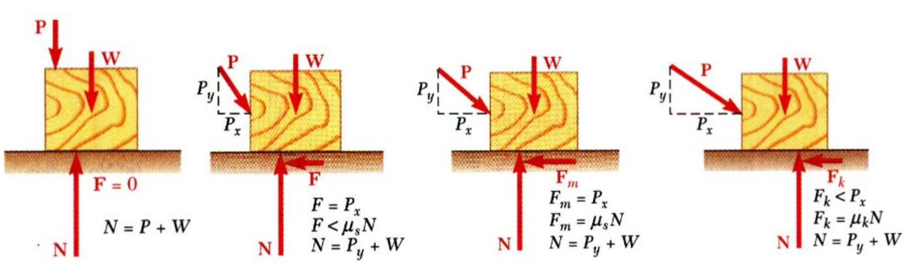
- Maximum static-friction force and kinetic-friction force are:
  - proportional to normal force
  - dependent on type and condition of contact surfaces
  - independent of contact area

When the surfaces are in relative motion, the contacts are more nearly along the tops of the humps, and the t-components of the *R*'s are smaller than when the surfaces are at rest relative to one another

→ Force necessary to maintain motion is generally less than that required to start the block when the surface irregularities are more nearly in mesh → $F_m > F_k$

Mechanism of Dry Friction

- Four situations can occur when a rigid body is in contact with a horizontal surface:



- No friction, ($P_x = 0$)

  Equations of Equilibrium Valid

- No motion, ($P_x < F_m$)

  Equations of Equilibrium Valid

- Motion impending, ($P_x = F_m$)

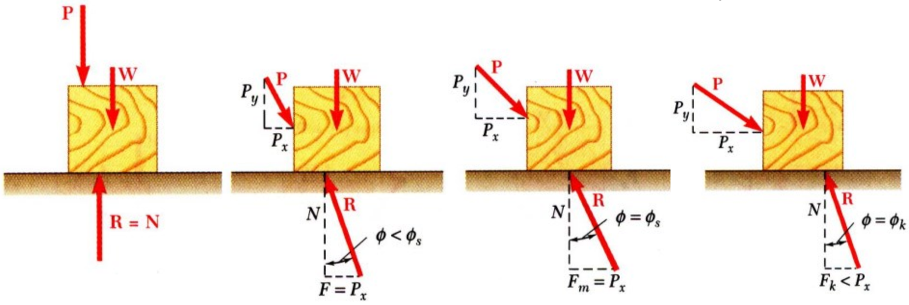
  Equations of Equilibrium Valid

- Motion, ($P_x > F_m$)

  Equations of Equilibrium Not Valid

Mechanism of Dry Friction

Sometimes convenient to replace normal force *N* & friction force *F* by their resultant ***R***:



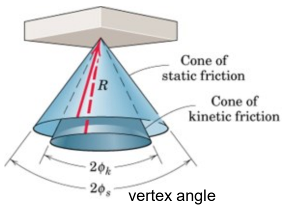
- No friction
- No motion
- Motion impending
- Motion

$$\tan\phi_s = \frac{F_m}{N} = \frac{\mu_s N}{N}$$

$$\tan\phi_s = \mu_s$$

$$\tan\phi_k = \frac{F_k}{N} = \frac{\mu_k N}{N}$$

$$\tan\phi_k = \mu_k$$

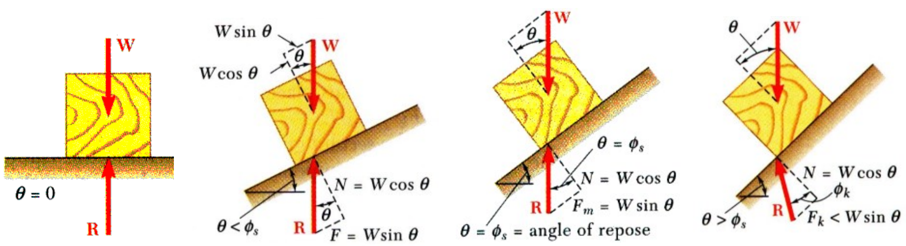


Friction Angles

ϕ_s = angle of static friction, ϕ_k = angle of kinetic friction

Mechanism of Dry Friction

- Consider block of weight *W* resting on board with variable inclination angle θ.



- No friction
- No motion
- Motion impending
- Motion

Angle of Repose = Angle of Static Friction

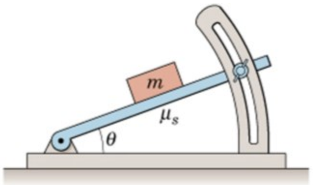
The reaction R is not vertical anymore, and the forces acting on the block are unbalanced

Dry Friction

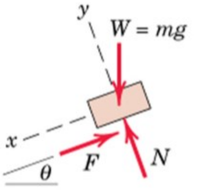
Example

Determine the maximum angle Θ before the block begins to slip.

μ_s = Coefficient of static friction between the block and the inclined surface



Solution: Draw the FBD of the block



$$[\Sigma F_x = 0] \qquad mg \sin\theta - F = 0 \qquad F = mg\sin\theta$$

$$[\Sigma F_y = 0] \qquad -mg\cos\theta + N = 0 \qquad N = mg\cos\theta$$

$$F/N = \tan\theta$$

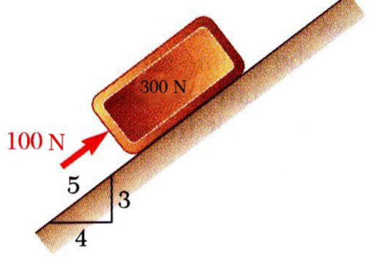
Max angle occurs when $F = F_{max} = \mu_s N$

Therefore, for impending motion: $\mu_s = \tan\theta_{max}$ or $\theta_{max} = \tan^{-1}\mu_s$

The maximum value of Θ is known as *Angle of Repose*

Dry Friction

Example

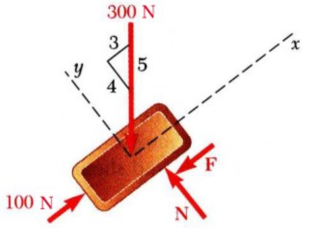


A 100 N force acts as shown on a 300 N block placed on an inclined plane. The coefficients of friction between the block and plane are μ_s = 0.25 and μ_k = 0.20. Determine whether the block is in equilibrium and find the value of the friction force.

SOLUTION:

- Determine values of friction force and normal reaction force from plane required to maintain equilibrium.
- Calculate maximum friction force and compare with friction force required for equilibrium. If it is greater, block will not slide.
- If maximum friction force is less than friction force required for equilibrium, block will slide. Calculate kinetic-friction force.

Dry Friction



SOLUTION:

- Determine values of friction force and normal reaction force from plane required to maintain equilibrium.

$$\sum F_x = 0: \quad 100\text{ N} - \tfrac{3}{5}(300\text{ N}) - F = 0$$

$$F = -80\text{ N} \quad \rightarrow F \text{ acting upwards}$$

$$\sum F_y = 0: \quad N - \tfrac{4}{5}(300\text{ N}) = 0$$

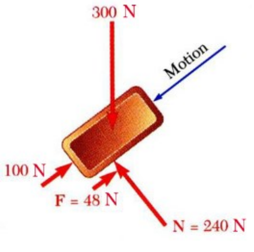
$$N = 240\text{ N}$$

- Calculate maximum friction force and compare with friction force required for equilibrium. If it is greater, block will not slide.

$$F_m = \mu_s N \qquad F_m = 0.25(240\text{ N}) = 60\text{ N}$$

The block will slide down the plane along F.

Dry Friction



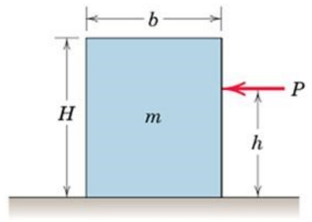
- If maximum friction force is less than friction force required for equilibrium, block will slide. Calculate kinetic-friction force.

$$F_{actual} = F_k = \mu_k \text{N}$$
$$= 0.20(240\text{ N})$$

$$\boxed{F_{actual} = 48\text{ N}}$$

Dry Friction

Example

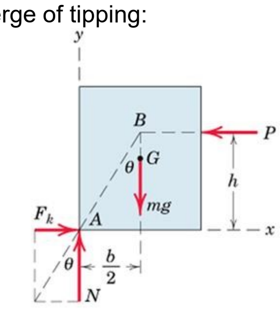


The block moves with constant velocity under the action of P. μ_k is the Coefficient of Kinetic Friction. Determine:

(a) Maximum value of h such that the block slides without tipping over

(b) Location of a point C on the bottom face of the block through which resultant of the friction and normal forces must pass if $h=H/2$

Solution: (a) FBD for the block on the verge of tipping:



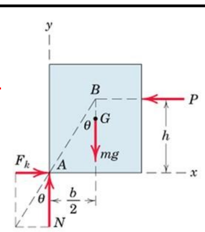
The resultant of F_k and N passes through point B through which P must also pass, since three coplanar forces in equilibrium are concurrent.

Friction Force:

$F_k = \mu_k N$ since slipping occurs

$\Theta = \tan^{-1}\mu_k$

Dry Friction



Solution (a) Apply Equilibrium Conditions (constant velocity!)

$$[\Sigma F_y = 0] \qquad N - mg = 0 \qquad N = mg$$

$$[\Sigma F_x = 0] \qquad F_k - P = 0 \qquad P = F_k = \mu_k N = \mu_k mg$$

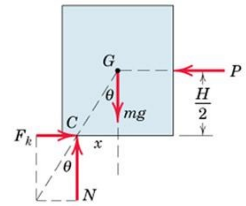
$$[\Sigma M_A = 0] \qquad Ph - mg\frac{b}{2} = 0 \qquad h = \frac{mgb}{2P} = \frac{mgb}{2\mu_k mg} = \frac{b}{2\mu_k}$$

Alternatively, we can directly write from the geometry of the FBD:

$$\tan\theta = \mu_k = \frac{b/2}{h} \qquad h = \frac{b}{2\mu_k}$$

If h were greater than this value, moment equilibrium at A would not be satisfied and the block would tip over.

Solution (b) Draw FBD



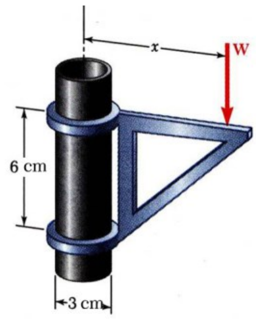
$\Theta = \tan^{-1}\mu_k$ since the block is slipping. From geometry of FBD:

$$\frac{x}{H/2} = \tan\theta = \mu_k \qquad \text{so} \qquad x = \mu_k H/2$$

Alternatively use equilibrium equations

Dry Friction

Example



The moveable bracket shown may be placed at any height on the 3-cm diameter pipe. If the coefficient of friction between the pipe and bracket is 0.25, determine the minimum distance *x* at which the load can be supported. Neglect the weight of the bracket.

SOLUTION:

- When W is placed at minimum x, the bracket is about to slip and friction forces in upper and lower collars are at maximum value.
- Apply conditions for static equilibrium to find minimum x.

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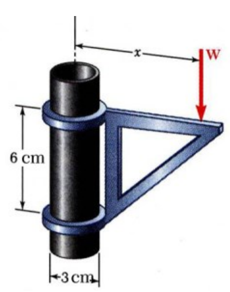
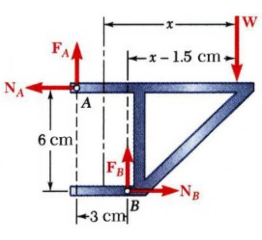
Dry Friction





SOLUTION:

- When W is placed at minimum x, the bracket is about to slip and friction forces in upper and lower collars are at maximum value.

$$F_A = \mu_s N_A = 0.25 N_A$$
$$F_B = \mu_s N_B = 0.25 N_B$$

- Apply conditions for static equilibrium to find minimum x.

$$\sum F_x = 0: \quad N_B - N_A = 0 \qquad N_B = N_A$$
$$\sum F_y = 0: \quad F_A + F_B - W = 0$$
$$0.25 N_A + 0.25 N_B - W = 0$$
$$0.5 N_A = W \qquad N_A = N_B = 2W$$
$$\sum M_B = 0: N_A(6\text{ cm}) - F_A(3\text{ cm}) - W(x - 1.5\text{ cm}) = 0$$
$$6N_A - 3(0.25 N_A) - W(x - 1.5) = 0$$
$$6(2W) - 0.75(2W) - W(x - 1.5) = 0$$
$$\boxed{x = 12\text{ cm}}$$

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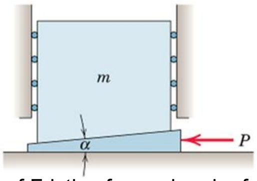
Applications of Friction in Machines

Wedges

- Simple machines used to raise heavy loads.
- Force required to lift block is significantly less than block weight.
- Friction prevents wedge from sliding out.
- Want to find minimum force P to raise block.



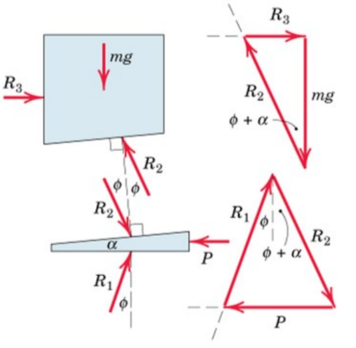
Coefficient of Friction for each pair of surfaces $\mu = \tan\phi$ (Static/Kinetic)

FBDs:
Reactions are inclined at an angle Φ from their respective normals and are in the direction opposite to the motion.

Force vectors acting on each body can also be shown.

R_2 is first found from upper diagram since mg is known.

Then P can be found out from the lower diagram since R_2 is known.



Forces to raise load

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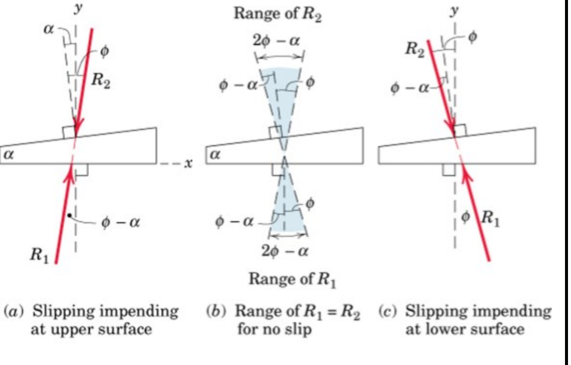
Applications of Friction in Machines: Wedges

P is removed and wedge remains in place
→ Equilibrium of wedge requires that the equal reactions R_1 and R_2 be collinear
→ In the figure, wedge angle α is taken to be less than ϕ

Impending slippage at the upper surface
Impending slippage at the lower surface

Slippage must occur at both surfaces simultaneously In order for the wedge to slide out of its space
→ Else, the wedge is Self-Locking

Range of angular positions of R_1 and R_2 for which the wedge will remain in place is shown in figure (b)



Simultaneous slippage is not possible if $\alpha < 2\phi$

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Applications of Friction in Machines: Wedges

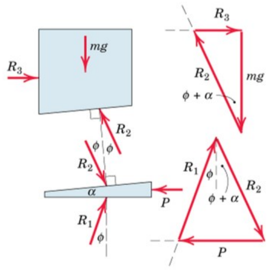
A pull P is required on the wedge for withdrawal of the wedge

→ **The reactions R_1 and R_2 must act on the opposite sides of their normal from those when the wedge was inserted**

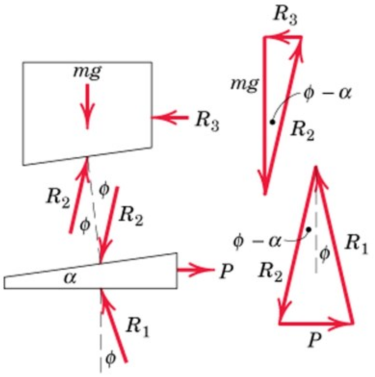
→ Solution by drawing FBDs and vector polygons

→ Graphical solution

→ Algebraic solutions from trigonometry



Forces to raise load



Forces to lower load

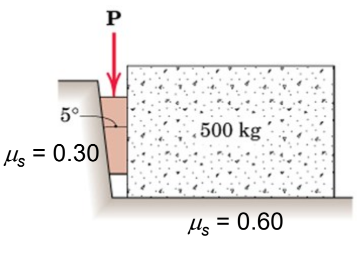
Applications of Friction in Machines

Example: Wedge

Coefficient of Static Friction for both pairs of wedge = 0.3

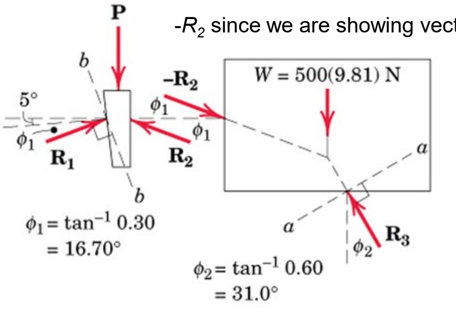
Coefficient of Static Friction between block and horizontal surface = 0.6

Find the least P required to move the block



Solution: Draw FBDs

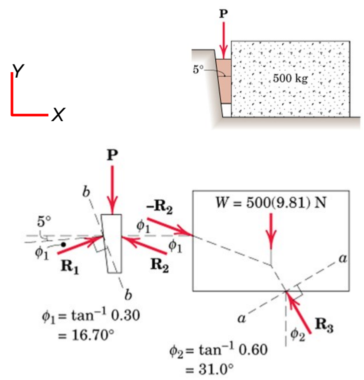
$-R_2$ since we are showing vectors



Applications of Friction in Machines

Solution: W = 500x9.81 = 4905 N

Three ways to solve



Method 1:

Equilibrium of FBD of the Block

$\sum F_X = 0$

$R_2 \cos \phi_1 = R_3 \sin \phi_2 \rightarrow R_2 = 0.538R_3$

$\sum F_Y = 0$

$4905 + R_2 \sin \phi_1 = R_3 \cos \phi_2 \rightarrow R_3 = 6970$ N

$\rightarrow R_2 = 3750$ N

Equilibrium of FBD of the Wedge

$\sum F_X = 0$

$R_2 \cos \phi_1 = R_1 \cos(\phi_1+5) \rightarrow R_1 = 3871$ N

$\sum F_Y = 0$

$R_1 \sin(\phi_1+5) + R_2 \sin \phi_1 = P$

$\rightarrow P = 2500$ N

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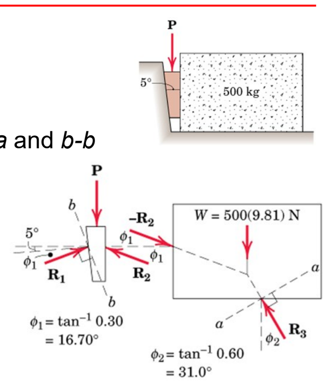
Applications of Friction in Machines

Solution:



Method 2:

Using Equilibrium equations along reference axes *a-a* and *b-b*

→ No need to solve simultaneous equations

Angle between R_2 and *a-a* axis = 16.70+31.0 = 47.7°

Equilibrium of Block:

$$[\Sigma F_a = 0] \quad 500(9.81) \sin 31.0° - R_2 \cos 47.7° = 0$$
$$R_2 = 3750 \text{ N}$$

Equilibrium of Wedge:

Angle between R_2 and *b-b* axis = $90-(2\Phi_1+5) = 51.6°$

Angle between P and *b-b* axis = $\Phi_1+5 = 21.7°$

$$[\Sigma F_b = 0] \quad 3750 \cos 51.6° - P \cos 21.7° = 0$$
$$P = 2500 \text{ N}$$

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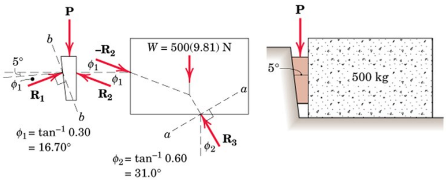
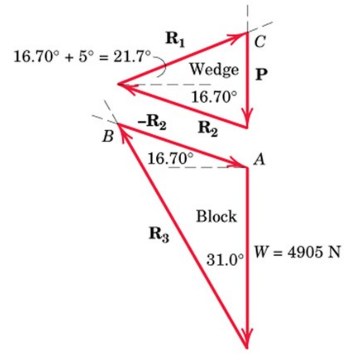
Applications of Friction in Machines

Solution:

Method 3:
Graphical solution using vector polygons



Starting with equilibrium of the block:
W is known, and directions of R_2 and R_3 are known
→ Magnitudes of R_2 and R_3 can be determined graphically

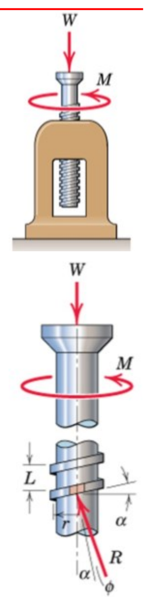


Similarly, construct vector polygon for the wedge from known magnitude of R_2, and known directions of R_2, R_1, *and P.*
→ Find out the magnitude of P graphically

Applications of Friction in Machines

Square Threaded Screws

- Used for fastening and for transmitting power or motion
- Square threads are more efficient
- Friction developed in the threads largely determines the action of the screw



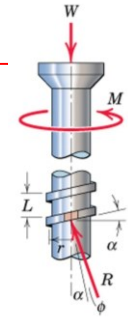
FBD of the Screw: R exerted by the thread of the jack frame on a small portion of the screw thread is shown

Lead = L = advancement per revolution
L = Pitch – for single threaded screw
L = 2xPitch – for double threaded screw (twice advancement per revolution)
Pitch = axial distance between adjacent threads on a helix or screw
Mean Radius = r ; α = Helix Angle

Similar reactions exist on all segments of the screw threads

Analysis similar to block on inclined plane since friction force does not depend on area of contact.
- Thread of base can be "unwrapped" and shown as straight line. Slope is $2\pi r$ horizontally and lead L vertically.

Applications of Friction in Machines: Screws

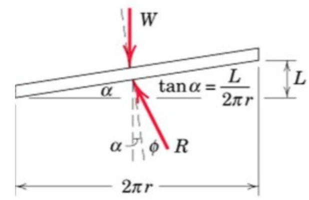


If M is just sufficient to turn the screw → Motion Impending
Angle of friction = ϕ (made by R with the axis normal to the thread)
→ $\tan \phi = \mu$

Moment of R @ vertical axis of screw = $R\sin(\alpha+\phi)r$
→ Total moment due to all reactions on the thread = $\sum R\sin(\alpha+\phi)r$
→ Moment Equilibrium Equation for the screw:
→ $M = [r \sin(\alpha + \phi)] \sum R$

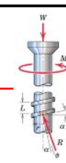
Equilibrium of forces in the axial direction: $W = \sum R \cos(\alpha + \phi)$
→ $W = [\cos(\alpha + \phi)] \sum R$

Finally → $\boldsymbol{M = W r \tan(\alpha + \phi)}$

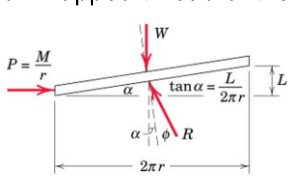


Helix angle α can be determined by unwrapping the thread of the screw for one complete turn
$\boldsymbol{\alpha = \tan^{-1} (L/2\pi r)}$

Applications of Friction in Machines: Screws

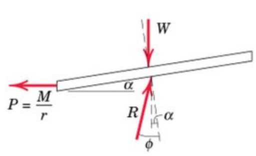


Alternatively, action of the entire screw can be simulated using unwrapped thread of the screw



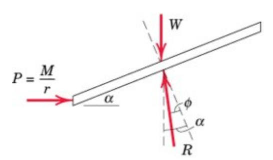
To Raise Load

Equivalent force required to push the movable thread up the fixed incline is:
$P = M/r$
From Equilibrium:
$M = W r \tan(\alpha + \phi)$
If M is removed: the screw will remain in place and be self-locking provided $\alpha<\phi$ and will be on the verge of unwinding if $\alpha=\phi$



To Lower Load ($\alpha<\phi$)

To lower the load by unwinding the screw, We must reverse the direction of M as long as $\alpha<\phi$
From Equilibrium:
$M = W r \tan(\phi - \alpha)$
→ This is the moment required to unwind the screw



To Lower Load ($\alpha>\phi$)

If $\alpha > \phi$, the screw will unwind by itself. Moment required to prevent unwinding:
From Equilibrium:
$M = W r \tan(\alpha - \phi)$

Applications of Friction in Machines

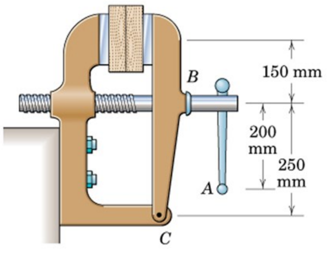
Example: Screw

Single threaded screw of the vise has a mean diameter of 25 mm and a lead of 5 mm. A 300 N pull applied normal to the handle at *A* produces a clamping force of 5 kN between the jaws of the vise. Determine:

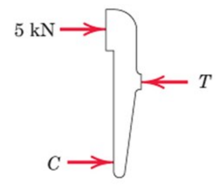
(a) Frictional moment M_B developed at *B* due to thrust of the screw against body of the jaw

(b) Force *Q* applied normal to the handle at *A* required to loosen the vise

μ_s in the threads = 0.20



Solution: Draw FBD of the jaw to find tension in the screw



$\sum M_C = 0$
$\rightarrow T = 8$ kN

Find the helix angle α and the friction angle ϕ

$\alpha = tan^{-1}(L/2\pi r) = 3.64°$

$\tan\phi = \mu \rightarrow \phi = 11.31°$

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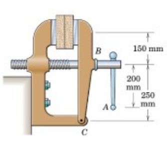
Applications of Friction in Machines

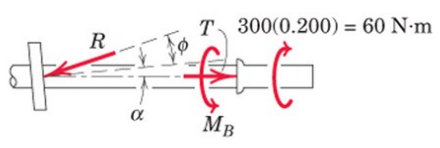


Example: Screw

Solution:

(a) To tighten the vise

Draw FBD of the screw



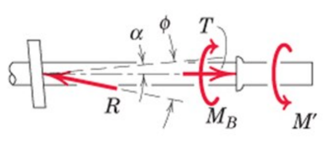
$M = T\,r\tan(\alpha + \phi)$

$60 - M_B = 8000(0.0125)\tan(3.64+11.31)$

$\boldsymbol{M_B = 33.3}$ **Nm**

(a) To loosen the vise (on the verge of being loosened)

Draw FBD of the screw: Net moment = applied moment *M'* minus M_B



$M = T\,r\tan(\phi - \alpha)$

$M' - 33.3 = 8000(0.0125)\tan(11.31-3.64)$

$M' = 46.8$ Nm

$\boldsymbol{Q = M'/d = 46.8/0.2 = 234}$ **N**

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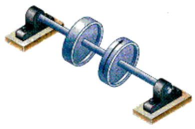
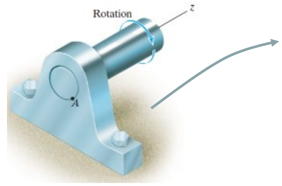
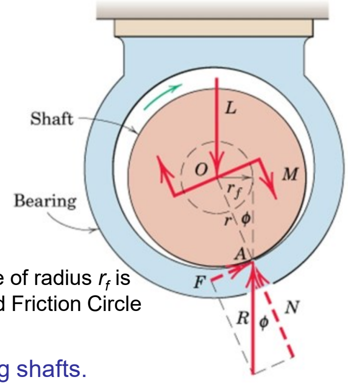
Applications of Friction in Machines

Journal Bearings (Axle Friction)







Circle of radius r_f is called Friction Circle

- Journal bearings provide lateral support to rotating shafts.
- Lateral load acting on the shaft is *L*.
- Thrust bearings provide axial support to rotating shafts.
- Frictional resistance of fully lubricated bearings depends on clearances, speed and lubricant viscosity.
- Partially lubricated axles and bearings can be assumed to be in direct contact along a straight line

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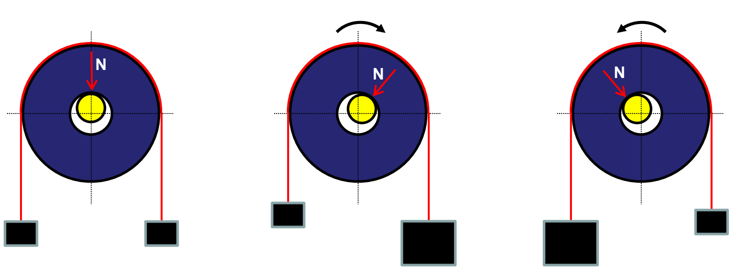
Applications of Friction in Machines

Journal Bearings (Axle Friction)

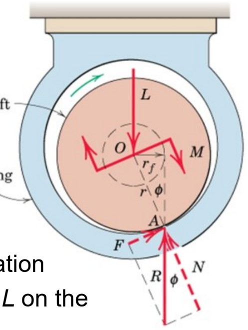
Exaggerated Figures show point of application of the normal reactions.

The frictional force will act normal to N and opposing the motion.

Resultant of frictional and normal force will act at an angle ϕ from N



Applications of Friction in Machines



Journal Bearings (Axle Friction)

Consider a dry or partially lubricated Journal Bearing

- with contact with near contact bet^n shaft and bearing
- As the shaft begins to turn in the direction shown, it will roll up the inner surface of bearing until it slips at A
- Shaft will remain in a more or less fixed position during rotation
- Torque *M* required to maintain rotation, and the radial load *L* on the shaft will cause reaction *R* at the contact point *A*.
- For vertical equilibrium, *R* must be equal to *L* but will not be collinear
- *R* will be tangent to a small circle of radius r_f called the *friction circle*

$\sum M_A = 0 \rightarrow M = Lr_f = Lr \sin\phi$

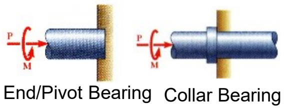
For a small coefficient of friction, ϕ is small → $\sin\phi \approx \tan\phi$

→ $M = \mu Lr$ (since $\mu = \tan\phi$) → Use equilibrium equations to solve a problem

→ Moment that must be applied to the shaft to overcome friction for a dry or partially lubricated journal bearing

Applications of Friction in Machines

Thrust Bearings (Disk Friction)



End/Pivot Bearing   Collar Bearing

- Thrust bearings provide axial support to rotating shafts.
- Axial load acting on the shaft is *P*.
- Friction between circular surfaces under distributed normal pressure (Ex: clutch plates, disc brakes)

Sanding Machine

Consider two flat circular discs whose shafts are mounted in bearings: they can be brought under contact under *P*

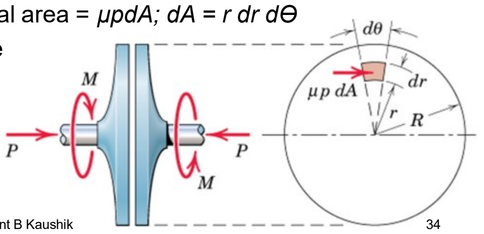
Max torque that the clutch can transmit = *M* required to slip one disc against the other

p is the normal pressure at any location between the plates

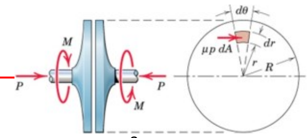
→ Frictional force acting on an elemental area = $\mu p dA$; $dA = r\,dr\,d\Theta$

Moment of this elemental frictional force about the shaft axis = $\mu p r dA$

Total $M = \int \mu p r dA$ over the area of disc



Applications of Friction in Machines



Thrust Bearings (Disk Friction)

Assuming that μ and p are uniform over the entire surface → $P = \pi R^2 p$

→ Substituting the constant p in $M = \int \mu prdA$ → $M = \frac{\mu P}{\pi R^2}\int_0^{2\pi}\int_0^R r^2 dr\, d\theta$

→ $M = \frac{2}{3}\mu PR$ ➔ Magnitude of moment reqd for impending rotation of shaft

≈ moment due to frictional force μp acting a distance ⅔ R from shaft center

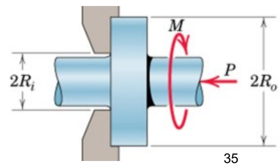
Frictional moment for worn-in plates is only about ¾ of that for the new surfaces
→ M for worn-in plates = ½(μPR)

If the friction discs are rings (Ex: Collar bearings) with outside and inside radius as R_o and R_i, respectively (limits of integration R_o and R_i) → $P = \pi(R_o^2 - R_i^2)p$

→ The frictional torque: $M = \frac{\mu P}{\pi(R_o^2 - R_i^2)}\int_0^{2\pi}\int_{R_i}^{R_o} r^2 dr\, d\theta$

→ $M = \frac{2}{3}\mu P \frac{R_o^3 - R_i^3}{R_o^2 - R_i^2}$

Frictional moment for worn-in plates → $M = \frac{1}{2}\mu P(R_o + R_i)$



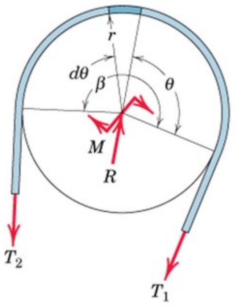
Applications of Friction in Machines

Belt Friction

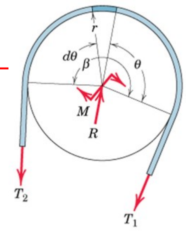
Impending slippage of flexible cables, belts, ropes over sheaves, wheels, drums
→ It is necessary to estimate the frictional forces developed between the belt and its contacting surface.

Consider a drum subjected to two belt tensions (T_1 and T_2)
M is the torque necessary to prevent rotation of the drum
R is the bearing reaction
r is the radius of the drum
β is the total contact angle between belt and surface
(β in radians)

$T_2 > T_1$ since M is clockwise



Applications of Friction in Machines



Belt Friction: Relate T_1 and T_2 when belt is about to slide to left

Draw FBD of an element of the belt of length $r\,d\theta$

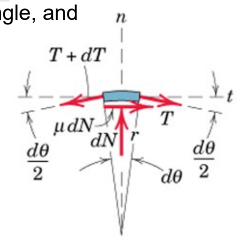
Frictional force for impending motion = $\mu\, dN$

Equilibrium in the *t*-direction: $T\cos\dfrac{d\theta}{2} + \mu\, dN = (T + dT)\cos\dfrac{d\theta}{2}$

→ $\mu dN = dT$ (cosine of a differential quantity is unity in the limit)

Equilibrium in the *n*-direction: $dN = (T + dT)\sin\dfrac{d\theta}{2} + T\sin\dfrac{d\theta}{2}$

→ $dN = 2Td\theta/2 = Td\theta$ (sine of a differential in the limit equals the angle, and product of two differentials can be neglected)



Combining two equations: $\dfrac{dT}{T} = \mu\, d\theta$

Integrating between corresponding limits: $\displaystyle\int_{T_1}^{T_2}\frac{dT}{T} = \int_0^{\beta}\mu\, d\theta$

→ $\ln\dfrac{T_2}{T_1} = \mu\beta$ → $T_2 = T_1\, e^{\mu\beta}$ ($T_2 > T_1$; e = 2.718…; β in radians)

- Rope wrapped around a drum *n* times → $\beta = 2\pi n$ radians
- *r* not present in the above eqn → eqn valid for non-circular sections as well
- In belt drives, belt and pulley rotate at constant speed → the eqn describes condition of impending slippage.

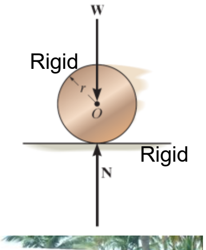
Applications of Friction in Machines

Wheel Friction or Rolling Resistance

Resistance of a wheel to roll over a surface is caused by deformation between two materials of contact.

→ This resistance is not due to tangential frictional forces

→ Entirely different phenomenon from that of dry friction



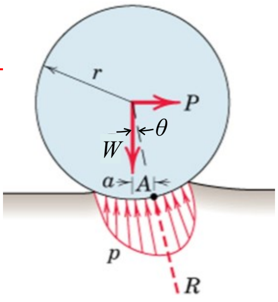
If a rigid cylinder rolls at constant velocity along a rigid surface, the normal force exerted by the surface on the cylinder acts at the tangent point of contact → No Rolling Resistance

Steel is very stiff
→ Low Rolling Resistance

Significant Rolling Resistance between rubber tyre and tar road

Large Rolling Resistance due to wet field

Applications of Friction in Machines



Wheel Friction or Rolling Resistance

Actually materials are not rigid → deformation occurs
→ reaction of surface on the cylinder consists of a distribution of normal pressure.

Consider a wheel under action of a load *W* on axle and a force *P* applied at its center to produce rolling
→ Deformation of wheel and supporting surface
→ Resultant *R* of the distribution of normal pressure must pass through wheel center for the wheel to be in equilibrium (i.e., rolling at a constant speed)
→ *R* acts at point *A* on right of wheel center for rightwards motion

Force *P* reqd to maintain rolling at constant speed can be appx estimated as:

$\sum M_A = 0$ → $Wa = Pr\cos\theta$ ($\cos\theta \approx 1$ ← deformations are very small compared to r)

→ $P = \dfrac{a}{r}W = \mu_r W$ μ_r is called the Coefficient of Rolling Resistance

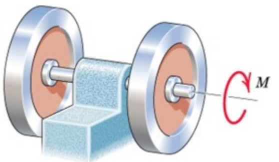
- μ_r is the ratio of resisting force to the normal force → analogous to μ_s or μ_k
- No slippage or impending slippage in interpretation of μ_r

Applications of Friction in Machines

Examples: Journal Bearings

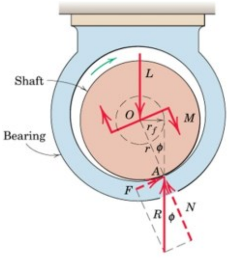
Two flywheels (each of mass 40 kg and diameter 40 mm) are mounted on a shaft, which is supported by a journal bearing. M = 3 Nm couple is reqd on the shaft to maintain rotation of the flywheels and shaft at a constant low speed.
Determine: (a) coeff of friction in the bearing, and (b) radius r_f of the friction circle.

Solution: Draw the FBD of the shaft and the bearing



(a) Moment equilibrium at O
$M = Rr_f = R r\sin\phi$
M = 3 Nm, R = 2x40x9.81 = 784.8 N, r = 0.020 m
→ $\sin\phi = 0.1911$ → $\phi = 11.02^{\circ}$

(b) $r_f = r\sin\phi$ = 3.82 mm



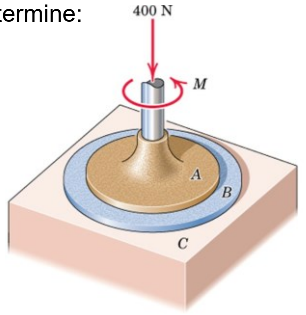
Applications of Friction in Machines

Examples: Disk Friction

Circular disk *A* (225 mm dia) is placed on top of disk *B* (300 mm dia) and is subjected to a compressive force of 400 N. Pressure under each disk is constant over its surface. Coeff of friction betn *A* and *B* = 0.4. Determine:

(a) the couple *M* which will cause *A* to slip on *B*.

(b) Min coeff of friction μ between *B* and supporting surface *C* which will prevent *B* from rotating.

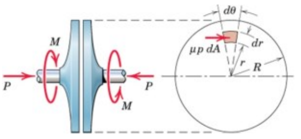


Solution: $M = \frac{2}{3}\mu P R$

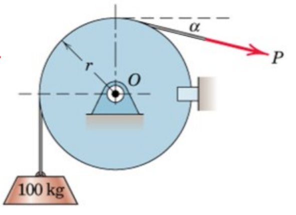
(a) Impending slip between *A* and *B*:
μ=0.4, *P*=400 N, R=225/2 mm
M = 2/3 x 0.4 x 400 x 0.225/2 → *M* = 12 Nm

(b) Impending slip between *B* and *C* :
Slip between *A* and *B* → *M* = 12 Nm

μ=? *P*=400 N, R=300/2 mm
12 = 2/3 x μ x 400 x 0.300/2 → μ = 0.3



Applications of Friction in Machines



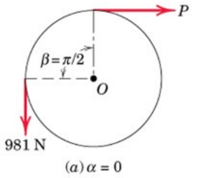
Examples: Belt Friction

A force *P* is reqd to be applied on a flexible cable that supports 100 kg load using a **fixed** circular drum.
μ between cable and drum = 0.3

(a) For α = 0, determine the max and min *P* in order not to raise or lower the load

(b) For *P* = 500 N, find the min α before the load begins to slip

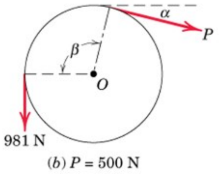
Solution: Impending slippage of the cable over the fixed drum is given by: $T_2 = T_1 e^{\mu\beta}$
Draw the FBD for each case



(a) α = 0

(a) μ = 0.3, α = 0, $\beta = \pi/2$ rad
For impending upward motion of the load: $T_2 = P_{max}$; T_1 = 981 N
$P_{max}/981 = e^{0.3(\pi/2)}$ → P_{max} = 1572 N
For impending downward motion: T_2 = 981 N; $T_1 = P_{min}$
$981/P_{min} = e^{0.3(\pi/2)}$ → P_{min} = 612 N

(b) μ = 0.3, α = ?, $\beta = \pi/2+\alpha$ rad, T_2 = 981 N; T_1 = 500 N
$981/500 = e^{0.3\beta}$ → $0.3\beta = \ln(981/500)$ → β = 2.25 rad
→ $\beta = 2.25\mathrm{x}(360/2\pi) = 128.7^{\circ}$
→ $\alpha = 128.7 - 90 = 38.7^{\circ}$

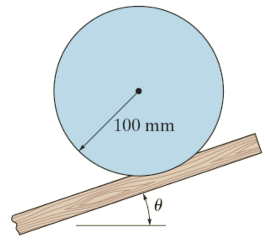


(b) P = 500 N

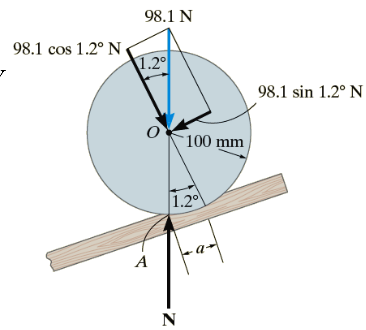
Applications of Friction in Machines

Examples: Rolling Resistance

A 10 kg steel wheel (radius = 100 mm) rests on an incline plane made of wood. At θ=1.2°, the wheel begins to roll-c the incline with constant velocity.
Determine the coefficient of rolling resistance.



Solution: When the wheel has impending motion, the normal reaction *N* acts at point *A* defined by the dimension *a*.
Draw the FBD for the wheel:
r = 100 mm, 10 kg = 98.1 N



Using simplified equation directly: $P = \frac{a}{r} W = \mu_r W$
Here *P* = 98.1(sin1.2) = 2.05 N
W = 98.1(cos1.2) = 98.08 N
→ Coeff of Rolling Resistance μ_r = 0.0209

Alternatively, $\sum M_A = 0$
→ 98.1(sin1.2)(*r* appx) = 98.1(cos1.2)*a*
(since *r*cos1.2 = *r*x0.9998 ≈ *r*)
→ $a/r = \mu_r = 0.0209$

Virtual Work

Method of Virtual Work

- Previous methods (FBD, ∑*F*, ∑*M*) are generally employed for a body whose equilibrium position is known or specified
- For problems in which bodies are composed of interconnected members that can move relative to each other.
- → various equilibrium configurations are possible and must be examined.
- → previous methods can still be used but are not the direct and convenient.
- **Method of Virtual Work** is suitable for analysis of multi-link structures (pin-jointed members) which change configuration
- effective when a simple relation can be found among the disp. of the pts of application of various forces involved
- → based on the concept of work done by a force
- → enables us to examine stability of systems in equilibrium


Scissor Lift Platform

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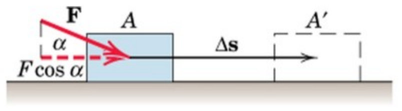
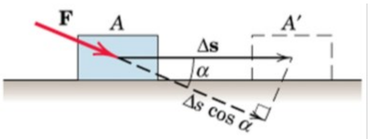
Virtual Work

Work done by a Force (U)

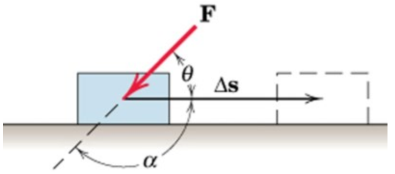




U = work done by the component of the force in the direction of the displacement times the displacement

$U = (F \cos \alpha)\ \Delta s$ or $U = F(\Delta s \cos \alpha)$

Since same results are obtained irrespective of the direction in which we resolve the vectors → Work is a scalar quantity



+U → Force and Disp in same direction

- U → Force and Disp in opposite direction

$$U = (F \cos \alpha)\ \Delta s = -(F \cos \theta)\ \Delta s$$

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Virtual Work

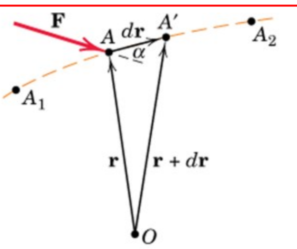
Work done by a Force (U)

Generalized Definition of Work

Work done by **F** during displacement $d\mathbf{r}$

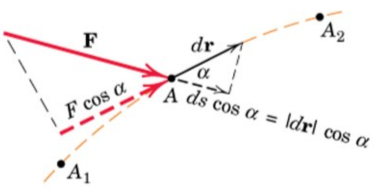
$$dU = \mathbf{F} \cdot d\mathbf{r}$$

→ $dU = F\ ds \cos \alpha$



Expressing **F** and $d\mathbf{r}$ in terms of their rectangular components

$$dU = (\mathbf{i}F_x + \mathbf{j}F_y + \mathbf{k}F_z) \cdot (\mathbf{i}\,dx + \mathbf{j}\,dy + \mathbf{k}\,dz)$$
$$= F_x\,dx + F_y\,dy + F_z\,dz$$



Total work done by **F** from A_1 to A_2

$$U = \int \mathbf{F} \cdot d\mathbf{r} = \int (F_x\,dx + F_y\,dy + F_z\,dz) \rightarrow U = \int F \cos \alpha\ ds$$

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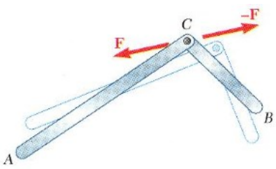
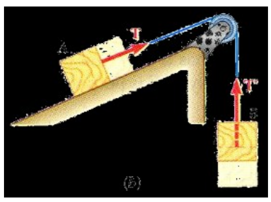
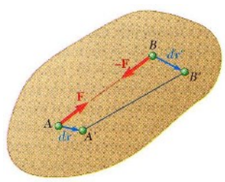
Virtual Work: Work done by a Force







Forces which do no work:

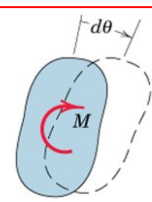
- forces applied to fixed points ($ds = 0$)
- forces acting in a dirn normal to the disp ($\cos\alpha = 0$)
- reaction at a frictionless pin due to rotation of a body around the pin
- reaction at a frictionless surface due to motion of a body along the surface
- weight of a body with cg moving horizontally
- friction force on a wheel moving without slipping

Sum of work done by several forces may be zero:

- bodies connected by a frictionless pin
- bodies connected by an inextensible cord
- internal forces holding together parts of a rigid body

Virtual Work

Work done by a Couple (*U*)

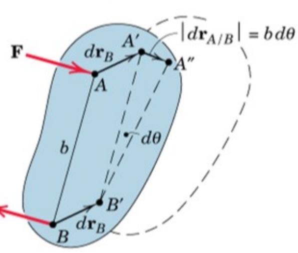


Small rotation of a rigid body:

• translation to *A'B'*

→ work done by *F* during disp *AA'* will be equal and opposite to work done by *-F* during disp *BB'*

→ total work done is zero



• rotation of *A'* about *B'* to *A"*

→ work done by *F* during disp *AA"* :

$U = \mathbf{F}.d\mathbf{r}_{A/B} = Fbd\theta$

Since $M = Fb$

→ $\boxed{dU = M\,d\theta}$ $+M$ → M has same sense as θ; $-M$ → M has opp sense as θ

Total word done by a couple during a finite rotation in its plane:

$$U = \int M\,d\theta$$

Virtual Work

Dimensions and Units of Work

(Force) x (Distance) → Joule (J) = N.m

→ Work done by a force of 1 Newton moving through a distance of 1 m in the direction of the force

→ Dimensions of Work of a Force and Moment of a Force are same though they are entirely different physical quantities.

→ **Work** is a scalar given by dot product; involves product of a force and distance, both measured along the same line

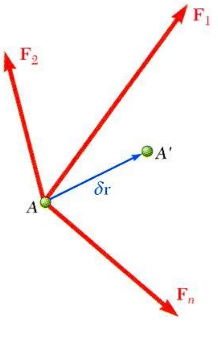
→ **Moment** is a vector given by the cross product; involves product of a force and distance measured at right angles to the force

→ Units of Work: Joule

→ Units of Moment: N.m

Virtual Work

Virtual Work: Disp does not really exist but only is assumed to exist so that we may compare various possible equilibrium positions to determine the correct one.



- *Imagine* the small *virtual displacement* of particle which is acted upon by several forces.
- The corresponding *virtual work*,

$$\delta U = \vec{F}_1 \cdot \delta\vec{r} + \vec{F}_2 \cdot \delta\vec{r} + \vec{F}_3 \cdot \delta\vec{r} = \left(\vec{F}_1 + \vec{F}_2 + \vec{F}_3\right) \cdot \delta\vec{r}$$

$$= \vec{R} \cdot \delta\vec{r}$$

Principle of Virtual Work:

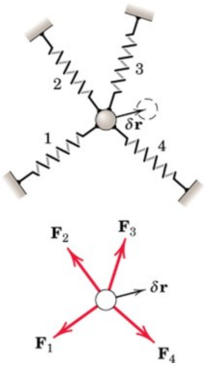
- If a particle is in equilibrium, the total virtual work of forces acting on the particle is zero for any virtual displacement.
- If a rigid body is in equilibrium, the total virtual work of external forces acting on the body is zero for any virtual displacement of the body.
- If a system of connected rigid bodies remains connected during the virtual displacement, only the work of the external forces need be considered since work done by internal forces (equal, opposite, and collinear) cancels each other.

Virtual Work

Equilibrium of a Particle

Total virtual work done on the particle due to virtual displacement $\delta \mathbf{r}$:

$$\delta U = \mathbf{F}_1 \cdot \delta \mathbf{r} + \mathbf{F}_2 \cdot \delta \mathbf{r} + \mathbf{F}_3 \cdot \delta \mathbf{r} + \cdots = \Sigma \mathbf{F} \cdot \delta \mathbf{r}$$



Expressing $\sum \mathbf{F}$ in terms of scalar sums and $\delta \mathbf{r}$ in terms of its component virtual displacements in the coordinate directions:

$$\delta U = \Sigma \mathbf{F} \cdot \delta \mathbf{r} = (\mathbf{i}\, \Sigma F_x + \mathbf{j}\, \Sigma F_y + \mathbf{k}\, \Sigma F_z) \cdot (\mathbf{i}\, \delta x + \mathbf{j}\, \delta y + \mathbf{k}\, \delta z)$$
$$= \Sigma F_x\, \delta x + \Sigma F_y\, \delta y + \Sigma F_z\, \delta z = 0$$

The sum is zero since $\sum \mathbf{F} = 0$, which gives $\sum F_x = 0$, $\sum F_y = 0$, $\sum F_z = 0$

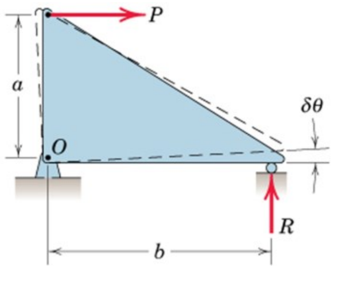
Alternative Statement of the equilibrium: $\delta U = 0$

This condition of zero virtual work for equilibrium is both necessary and sufficient since we can apply it to the three mutually perpendicular directions
→ **3 conditions of equilibrium**

Virtual Work: Applications of Principle of Virtual Work

Equilibrium of a Rigid Body

Total virtual work done on the entire rigid body is zero since virtual work done on each Particle of the body in equilibrium is zero.



Weight of the body is negligible.

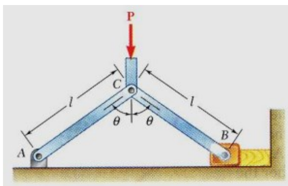
Work done by $P = -Pa\,\delta\theta$
Work done by $R = +Rb\,\delta\theta$

Principle of Virtual Work: $\delta U = 0$:
$-Pa\,\delta\theta + Rb\,\delta\theta = 0$

→ $Pa - Rb = 0$
→ Equation of Moment equilibrium @ O.
→ Nothing gained by using the Principle of Virtual Work for a single rigid body

Virtual Work: Applications of Principle of Virtual Work

Determine the force exerted by the vice on the block when a given force P is applied at C. Assume that there is no friction.



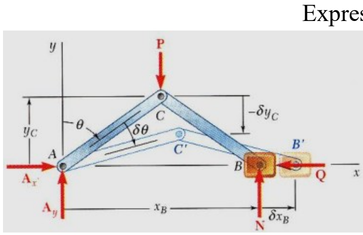
- Consider the work done by the external forces for a virtual displacement $\delta\theta$. $\delta\theta$ is a positive increment to θ in bottom figure. Only the forces P and Q produce nonzero work.
- x_B increases while y_C decreases

→ +ve increment for x_B: δx_B → $\delta U_Q = -Q\delta x_B$ (opp. Sense)

→ -ve increment for y_C: $-\delta y_C$ → $\delta U_P = +P(-\delta y_C)$ (same Sense)

$$\delta U = 0 = \delta U_Q + \delta U_P = -Q\,\delta x_B - P\,\delta y_C$$

Expressing x_B and y_C in terms of θ and differentiating w.r.t. θ



$$x_B = 2l\sin\theta \qquad y_C = l\cos\theta$$

$$\delta x_B = 2l\cos\theta\,\delta\theta \qquad \delta y_C = -l\sin\theta\,\delta\theta$$

$$0 = -2Ql\cos\theta\,\delta\theta + Pl\sin\theta\,\delta\theta$$

$$Q = \tfrac{1}{2}P\tan\theta$$

By using the method of virtual work, all unknown reactions were eliminated. $\sum M_A$ would eliminate only two reactions.

- If the virtual displacement is consistent with the constraints imposed by supports and connections, only the work of loads, applied forces, and friction forces need be considered.

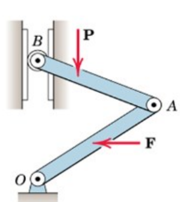
Virtual Work

Principle of Virtual Work

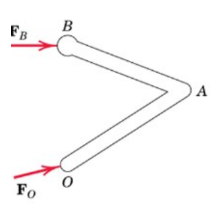
Virtual Work done by external active forces on an ideal mechanical system in equilibrium is zero for any and all virtual displacements consistent with the constraints

$$\delta U = 0$$

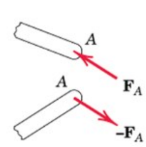
Three types of forces act on interconnected systems made of rigid members



Active Forces: Work Done
Active Force Diagram



Reactive Forces
No Work Done



Internal Forces
No Work Done

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Virtual Work

Major Advantages of the Virtual Work Method

- It is not necessary to dismember the systems in order to establish relations between the active forces.
- Relations between active forces can be determined directly without reference to the reactive forces.

→ The method is particularly useful in determining the position of equilibrium of a system under known loads (This is in contrast to determining the forces acting on a body whose equilibrium position is known – studied earlier).

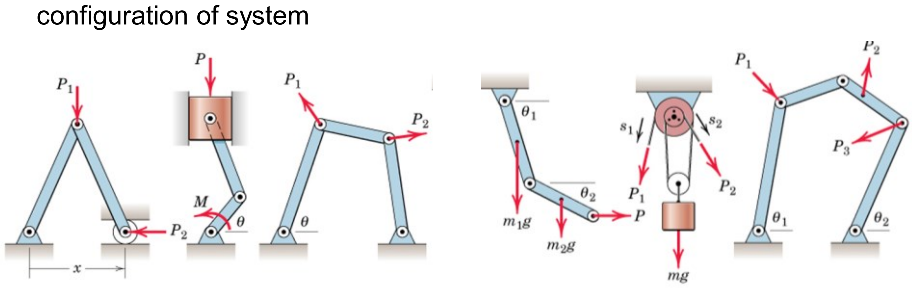
→ The method requires that internal frictional forces do negligible work during any virtual displacement.

→ If internal friction is appreciable, work done by internal frictional forces must be included in the analysis.

Virtual Work

Degrees of Freedom (DOF)

- Number of independent coordinates needed to specify completely the configuration of system



(*a*) Examples of one-degree-of-freedom systems

(*b*) Examples of two-degree-of-freedom systems

Only one coordinate (displacement or rotation) is needed to establish position of every part of the system

Two independent coordinates are needed to establish position of every part of the system

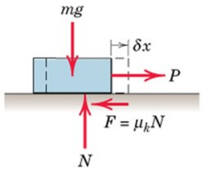
$\delta U=0$ can be applied to each DOF at a time keeping other DOF constant.

ME101 → only SDOF systems

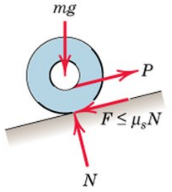
Virtual Work

Systems with Friction

- So far, the Principle of virtual work was discussed for "ideal" systems.
- If significant friction is present in the system ("Real" systems), work done by the external active forces (input work) will be opposed by the work done by the friction forces.



During a virtual displacement δx:
Work done by the kinetic friction force is: $-\mu_k N \delta x$



During rolling of a wheel:
the static friction force does no work if the wheel does not slip as it rolls.

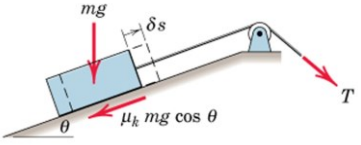
Virtual Work

Mechanical Efficiency (*e*)

- Output work of a machine is always less than the input work because of energy loss due to friction.

$$e = \frac{\text{Output Work}}{\text{Input Work}}$$

For simple machines with SDOF & which operates in uniform manner, mechanical efficiency may be determined using the method of Virtual Work



For the virtual displacement δs: Output Work is that necessary to elevate the block = $mg\ \delta s\ \sin\theta$

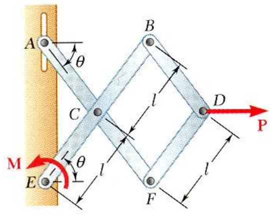
Input Work: $T\ \delta s = mg \sin\theta\ \delta s + \mu_k\ mg \cos\theta\ \delta s$
The efficiency of the inclined plane is:

$$e = \frac{mg\,\delta s \sin\theta}{mg(\sin\theta + \mu_k \cos\theta)\delta s} = \frac{1}{1 + \mu_k \cot\theta}$$

As friction decreases, Efficiency approaches unity

Virtual Work

Example



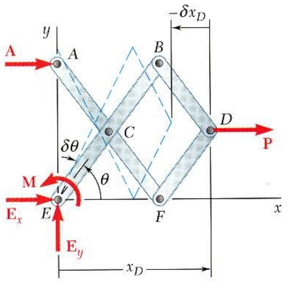
Determine the magnitude of the couple *M* required to maintain the equilibrium of the mechanism.

SOLUTION: Apply a positive virtual increment to θ at E

- Apply the principle of virtual work

$$\delta U = 0 = \delta U_M + \delta U_P$$
$$0 = M\delta\theta + P\delta x_D$$

$\delta\theta$ along positive θ → θ increases with $\delta\theta$ → + $\delta\theta$
M and $\delta\theta$ have same sense → + $M\delta\theta$
For positive $\delta\theta$, x_D will decrease by δx_D → - δx_D
P and - δx_D have opposite sense → - P(- δx_D) → + $P\delta x_D$



$$x_D = 3l\cos\theta$$
$$\delta x_D = -3l\sin\theta\delta\theta$$

$$0 = M\delta\theta + P(-3l\sin\theta\delta\theta)$$

$$\boxed{M = 3Pl\sin\theta}$$

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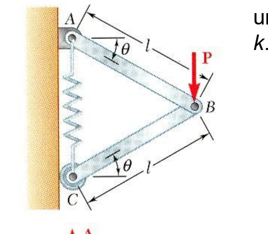
Virtual Work

Example

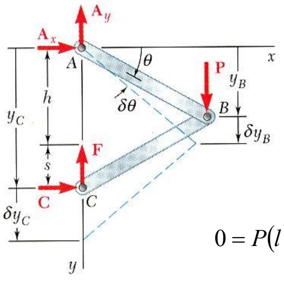


Determine the expressions for θ and for the tension in the spring which correspond to the equilibrium position of the spring. The unstretched length of the spring is *h* and the constant of the spring is *k*. Neglect the weight of the mechanism.

SOLUTION: Apply a positive virtual increment to θ at A

- Apply the principle of virtual work

$$\delta U = \delta U_B + \delta U_F = 0$$
$$0 = P\delta y_B - F\delta y_C$$



y_B increases with δy_B → + δy_B
P and δy_B have same sense→ + $P\ \delta y_B$
For positive $\delta\theta$, y_C will increase by δy_C → + δy_C
F and δy_C have opposite sense → -$F(\delta y_C)$ → - $F\delta y_C$

$$y_B = l\sin\theta \qquad y_C = 2l\sin\theta \qquad F = ks$$
$$\delta y_B = l\cos\theta\delta\theta \qquad \delta y_C = 2l\cos\theta\delta\theta \qquad = k(y_C - h)$$
$$= k(2l\sin\theta - h)$$

$$0 = P(l\cos\theta\delta\theta) - k(2l\sin\theta - h)(2l\cos\theta\delta\theta)$$

$$\boxed{\sin\theta = \frac{P + 2kh}{4kl} \qquad F = \tfrac{1}{2}P}$$

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Virtual Work

Potential Energy and Stability

- Till now equilibrium configurations of mechanical systems composed of rigid members was considered for analysis using method of virtual work.
- Extending the method of virtual work to account for mechanical systems which include elastic elements in the form of springs (non-rigid elements).
- Introducing the concept of Potential Energy, which will be used for determining the stability of equilibrium.
- Work done on an elastic member is stored in the member in the form of Elastic Potential Energy V_e.
    - This energy is potentially available to do work on some other body during the relief of its compression or extension.

Virtual Work

Elastic Potential Energy (V_e)

Consider a **linear and elastic** spring compressed by a force $F \rightarrow F = kx$
k = spring constant or stiffness of the spring

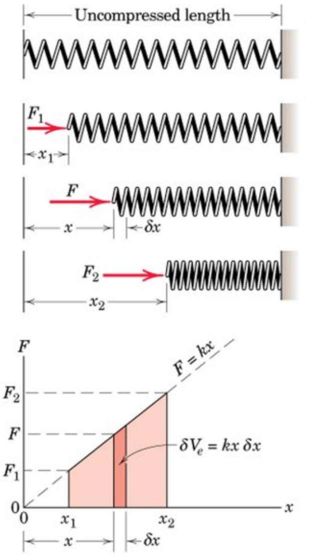
Work done on the spring by F during a movement dx: $dU = F\,dx$

Elastic potential energy of the spring for compression x = total work done on the spring

$$V_e = \int_0^x F\,dx = \int_0^x kx\,dx$$

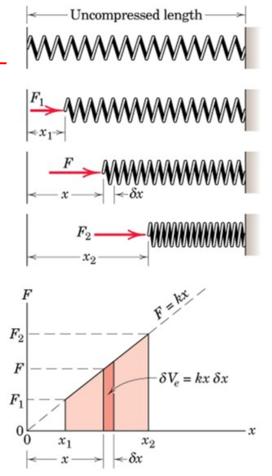
$$\boxed{V_e = \frac{1}{2}kx^2}$$

→ Potential Energy of the spring = area of the triangle in the diagram of F versus x from 0 to x



Virtual Work

Elastic Potential Energy



During increase in compression from x_1 to x_2:
Work done on the springs = change in V_e

$$\Delta V_e = \int_{x_1}^{x_2} kx\,dx = \tfrac{1}{2}k(x_2^2 - x_1^2)$$

→ Area of trapezoid from x_1 to x_2

During a virtual displacement δx of the spring:
virtual work done on the spring = virtual change in elastic potential energy

$$\delta V_e = F\,\delta x = kx\,\delta x$$

During the decrease in compression of the spring as it is relaxed from $x=x_2$ to $x=x_1$, the change (final minus initial) in the potential energy of the spring is negative
→ If δx is negative, δV_e is negative

When the spring is being stretched, the force acts in the direction of the displacement → Positive Work on the spring → Increase in the Potential Energy

Virtual Work

Elastic Potential Energy

Work done by the linear spring on the body to which the spring is attached (during displacement of the spring) is the negative of the change in the elastic potential energy of the spring (due to equilibrium).

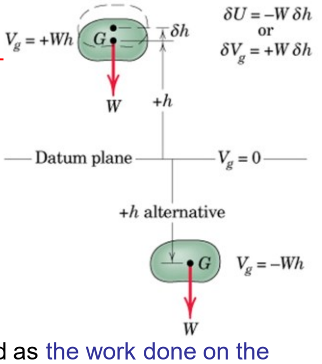
Torsional Spring: resists the rotation
K = Torsional Stiffness (torque per radian of twist)
θ = angle of twist in radians
Resisting torque, $M = K\theta$
The Potential Energy: $V_e = \int_0^\theta K\theta\,d\theta$ → $\boxed{V_e = \tfrac{1}{2}K\theta^2}$

This is similar to the expression for the linear extension spring

Units of Elastic Potential Energy → Joules (J)
(same as those of Work)

Virtual Work

Gravitational Potential Energy (V_g)



Treatment of Gravitational force till now:
For an upward displacement δh of the body, work done by the weight ($W=mg$) is: $\delta U = -mg\delta h$
For downward displacement (with h measured positive downward): $\delta U = mg\delta h$

Alternatively, work done by gravity can be expressed in terms of a change in potential energy of the body.
The Gravitational Potential Energy of a body is defined as the work done on the body by a force **equal and opposite to the weight** in bringing the body to the position under consideration from some arbitrary datum plane where the potential energy is defined to be zero. → V_g is negative of the work done by the weight.

V_g =0 at h=0
→ at height h above the datum plane, V_g of the body = mgh
→ If the body is at distance h below the datum plane, V_g of the body = $-mgh$

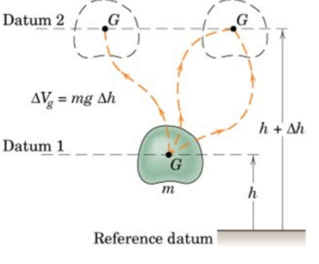
Virtual change in the gravitational potential energy: $\boldsymbol{\delta V_g = +mg\delta h}$
δh is the upward virtual displacement of the mass centre of the body
If mass centre has downward virtual displacement → $\boldsymbol{\delta V_g = -mg\delta h}$

Virtual Work

Gravitational Potential Energy (V_g)

Datum Plane for zero potential energy is arbitrary
- because only the change in potential energy matters
- the change remains the same irrespective of location of datum plane

V_g is also independent of the path followed in arriving at a particular level h.
- The body of mass m has same potential energy change in going from datum 1 to datum 2 (3 possible paths) because δh remains the same for all three paths.



Units of V_g are the same as those for the work and elastic potential energy: Joules (J)

Virtual Work

Energy Equation

Work done by the linear spring on the body to which the spring is attached (during displacement of the spring) is the negative of the change in the elastic potential energy of the spring.

Work done by the gravitational force or weight mg is the negative of the change in gravitational potential energy

Virtual Work eqn to a system with springs and with changes in the vertical position of its members → replace the work of the springs and the work of the weights by negative of the respective potential energy changes

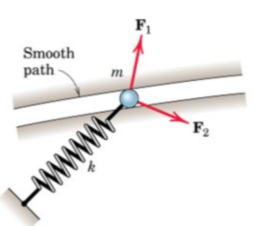
Total Virtual Work δU = work done by all active forces ($\delta U'$) other than spring forces and weight forces + the work done by the spring forces and weight forces, i.e., $-(\delta V_e + \delta V_g)$

$$\boxed{\delta U' - (\delta V_e + \delta V_g) = 0} \quad \text{or} \quad \boxed{\delta U' = \delta V}$$

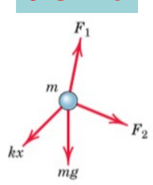
$V = V_e + V_g$ → Total Potential Energy of the system

Virtual Work

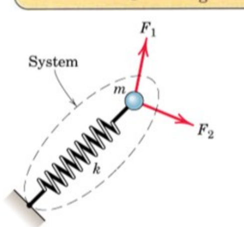
Active Force Diagrams: Use of two equations



$\boxed{\delta U = 0}$



$$\boxed{\delta U' - (\delta V_e + \delta V_g) = 0}$$

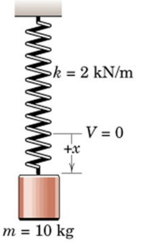


Restating the Principle of Virtual Work for a mechanical system with elastic members and members that undergo changes in position:

Virtual work done by all external active forces (other than the gravitational and spring forces accounted for in the potential energy terms) on a mechanical system in equilibrium = the corresponding change in the total elastic and gravitational potential energy of the system for any and all virtual displacements consistent with the constraints

Virtual Work

Stability of Equilibrium



Consider a system where movement is accompanied by changes in gravitational and elastic potential energy:
If work done by all active forces other than spring forces and weight forces is zero → $\delta U' = 0$ (No work is done on the system by the non-potential forces)

→ $\boxed{\delta(V_e + V_g) = 0}$ or $\boxed{\delta V = 0}$

→ Equilibrium configuration of a mechanical system is one for which the total potential energy V of the system has a **stationary value**.

For a SDOF system (where potential energy and its derivatives are continuous functions of a single variable, x that describes the configuration), it is equivalent to state that:

$$\boxed{\frac{dV}{dx} = 0}$$

→ A mechanical system is in equilibrium when the derivative of its total potential energy is zero

→ For systems with multiple DOF, partial derivative of V wrt each coordinate must be zero for equilibrium.

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$$\boxed{\frac{dV}{dx} = 0}$$

Stability of Equilibrium: SDOF

Three Conditions under which this eqn applies.
when total potential energy is:
→ A Minimum (Stable Equilibrium)
→ A Maximum (Unstable Equilibrium)
→ A Constant (Neutral Equilibrium)

Total Potential Energy: Minimum Maximum Constant

A small displacement away from the STABLE position results in an increase in potential energy and a tendency to return to the position of lower energy.

A small displacement away from the UNSTABLE position results in a decrease in potential energy and a tendency to move farther away from the equilibrium position to a position of still lower energy.

For a NEUTRAL position, a small displacement one way or the other results in no change in potential energy and no tendency to move either way.

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Stability of Equilibrium: SDOF systems

Evaluation of type of Stability at the point of Equilibrium:
When a function and its derivatives are continuous:
→ the second derivative is positive at a point of minimum value of function
→ the second derivative is negative at a point of maximum value of function

Mathematical conditions for equilibrium and stability of a system with a SDOF *x*:

Equilibrium: $\frac{dV}{dx} = 0$

Stable Equilibrium: $\frac{dV}{dx} = 0, \frac{d^2V}{dx^2} > 0$

Unstable Equilibrium: $\frac{dV}{dx} = 0, \frac{d^2V}{dx^2} < 0$

Neutral Equilibrium: $\frac{dV}{dx} = \frac{d^2V}{dx^2} = \frac{d^3V}{dx^3} = ... = 0$

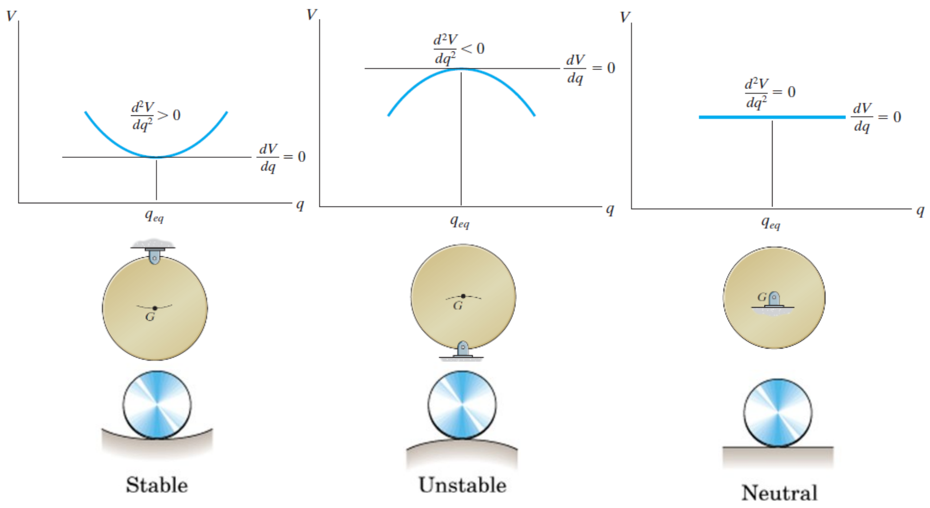
If $d^2V/dx^2 = 0$ at the equilibrium position, examine the sign of a higher derivative to ascertain the type of equilibrium.

- If the order of the lowest remaining non-zero derivative is even, the equilibrium will be stable or unstable according to the whether the sign of this derivative is positive or negative.
- If the order of the derivative is odd, the equilibrium is classified as unstable.

Stability criteria for MDOF systems require more advanced treatment.

Virtual Work: Stability of Equilibrium: SDOF systems

Evaluation of type of Stability at the point of Equilibrium:



V: Minimum Maximum Constant

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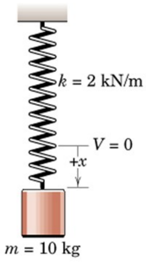


Example: A 10 kg cylinder is suspended by the spring. Plot the potential energy of the system and show that it is minimum at the equilibrium position.

Solution: **No external active forces**. Choosing datum plane for zero potential energy at the position where the spring is un-extended.

For an arbitrary position x: Elastic potential energy: $V_e = ½\, kx^2$

Gravitational Potential Energy: $V_g = -\, mgx$ (-ve of work done)

Total Potential Energy: $V = V_e + V_g = ½\, kx^2 - mgx$

Equilibrium occurs where $\boldsymbol{dV/dx = 0}$

→ $kx - mg = 0$ → $\boldsymbol{x = mg/k}$

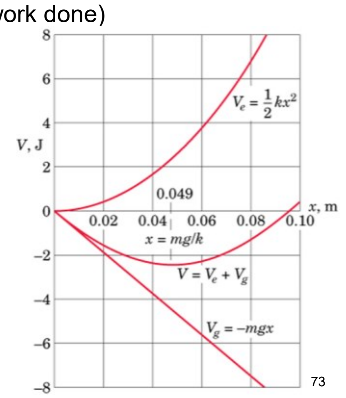
$d^2V/dx^2 = +\, k$ → Stable Equilibrium since positive

Substituting values of m and k

$V = ½\,(2000)x^2 - 10(9.81)x$

$x = 10(9.81)/2000 = 0.049$ m = 49 mm

Plot V vs x graph for various values of x



Minimum value of V occurs at x = 49 mm where $dV/dx = 0$ and d^2V/dx^2 is positive

Virtual Work: Stability of Equilibrium

Example: Both ends of the uniform bar of mass m slide freely along the guides. Examine the stability conditions for the position of equilibrium. The spring of stiffness k is un-deformed when x=0

Solution: The system consists of the spring and the bar

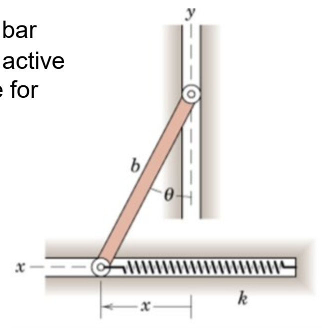
There are no external active forces. The Figure shows active force diagram. Choosing the x-axis as the datum plane for zero gravitational potential energy.

In the displaced position x:

$V_e = ½\, kx^2 = ½\, k\,(b^2\sin^2\theta)$

$V_g = mg\,(b/2)\cos\theta$ (-ve of work done)

$V = V_e + V_g = ½\, k\,(b^2\sin^2\theta) + mg\,(b/2)\cos\theta$



Equilibrium occurs for $\boldsymbol{dV/d\theta = 0}$

→ $kb^2\sin\theta\cos\theta - ½\, mgb\sin\theta = 0$

→ $(kb^2\cos\theta - ½\, mgb)\sin\theta = 0$ → Two Solutions

→ Two equilibrium positions at:

→ $\sin\theta = 0$ (and $\theta = 0$) and $\cos\theta = mg/(2kb)$

Determine the stability for each of the two equilibrium position by examining the sign of $d^2V/d\theta^2$

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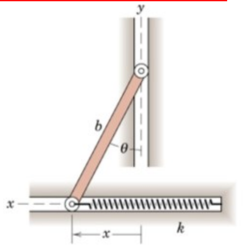
Example Solution:



$d^2V/d\theta^2 = kb^2(\cos^2\theta - \sin^2\theta) - ½\, mgb \cos\theta$
$\quad = kb^2(2\cos^2\theta - 1) - ½\, mgb \cos\theta$

Solution I: $\sin\theta = 0$ and $\theta = 0$
$d^2V/d\theta^2 = kb^2(1 - mg/2kb)$

If $mg/2kb < 1$, i.e., $mg/2b < k$ → $d^2V/d\theta^2$ is positive (Stable)
If $mg/2kb > 1$, i.e., $mg/2b > k$ → $d^2V/d\theta^2$ is negative (Unstable)
→ If the spring is sufficiently stiff, the bar will return to its original vertical position, even though there is no force in the spring at that position.

Solution II: $\cos\theta = mg/(2kb)$ and $\theta = \cos^{-1}(mg/2kb)$
$d^2V/d\theta^2 = kb^2 [(mg/2kb)^2 - 1]$
→ this will be always be negative (Unstable) because $\cos\theta$, i.e., $mg/(2kb)$, must be less than unity

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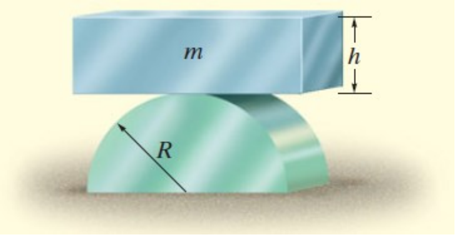
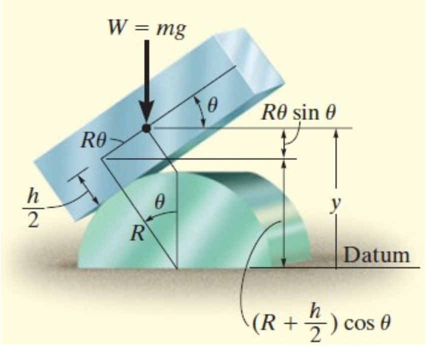
Virtual Work: Stability of Equilibrium

Example: A homogenous block of mass m rests on top surface of the cylinder. Show that this is a condition of unstable equilibrium if $h > 2R$





Solution: Choosing the base of the cylinder as the datum plane for zero gravitational potential energy.

$V = V_e + V_g = 0 + mgy$ (-ve of work done)

$$y = \left(R + \frac{h}{2}\right)\cos\theta + R\theta\sin\theta$$

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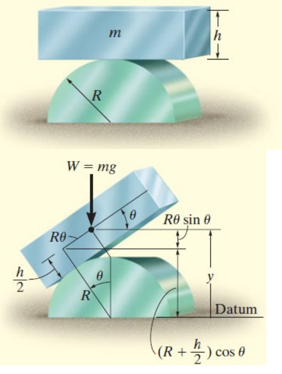
Example: Solution

$$V = mg\left[\left(R + \frac{h}{2}\right)\cos\theta + R\theta\sin\theta\right]$$



At equilibrium:

$$\frac{dV}{d\theta} = mg\left[-\left(R + \frac{h}{2}\right)\sin\theta + R\sin\theta + R\theta\cos\theta\right] = 0$$

$$= mg\left(-\frac{h}{2}\sin\theta + R\theta\cos\theta\right) = 0$$

$\theta = 0$ is the equilibrium position that satisfies this equation

Determine the stability of the system at $\theta = 0$ by examining the sign of $d^2V/d\theta^2$

$$\frac{d^2V}{d\theta^2} = mg\left(-\frac{h}{2}\cos\theta + R\cos\theta - R\theta\sin\theta\right) \qquad \left.\frac{d^2V}{d\theta^2}\right|_{\theta=0^\circ} = -mg\left(\frac{h}{2} - R\right)$$

Since all the constants are positive, the block is in unstable equilibrium if h/2 > R, i.e., h > 2R because then $d^2V/d\theta^2$ will be negative.

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